# Undetermined coefficients for nonhomogeneous equations

We said earlier that linear second order homogeneous equations with constant coefficients would take the form

$$y'' + Ay' + By = 0$$

Now we want to tackle nonhomogeneous equations, in which we're swapping out the zero-value on the right side for a non-zero value.

$$y'' + Ay' + By = g(x)$$

To solve nonhomogeneous equations like these, we'll first use the characteristic equation to find the general solution to the associated homogeneous equation, and then we'll deal with the $g(x)$ part in another step.

That being said, we'll be looking for the same kind of solutions to the homogeneous equation.

| **Roots** | **General solution** |
|---|---|
| Distinct real roots | $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$ |
| Equal real roots | $y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$ |
| Complex conjugate roots | $y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$ <br> where $r = \alpha \pm \beta i$ |

To address the $g(x)$ portion of the equation, there are multiple methods we can use, the first of which is the method of undetermined coefficients.

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## Undetermined coefficients

**Undetermined coefficients** is a method we can use to find the general solution $y(x)$ to a second order nonhomogeneous differential equation. The general solution to the nonhomogeneous equation will always be the sum of the complementary solution $y_c(x)$ and the particular solution $y_p(x)$.

$$y(x) = y_c(x) + y_p(x)$$

The **complementary solution** is the solution we find when we solve the characteristic equation (from the associated homogeneous equation). Which means we'll always start by pretending that the nonhomogeneous equation is actually a homogenous equation, replacing $g(x)$ with $0$.

Once we find the complementary solution, it's time to use $g(x)$ to make a guess about a **particular solution**, which is the solution that addresses the $g(x)$ portion of the differential equation. It takes practice to get good at building a particular solution, but here are some general guidelines.

- For a polynomial function like $x^2 + 1$, guess $Ax^2 + Bx + C$, making sure to include the highest-degree term, as well as all lower-degree terms. So if the highest degree term is $x^3$, we'll need to include $x^3$, $x^2$, $x$, and a constant.
- For an exponential function like $e^{3x}$, guess $Ae^{3x}$.
- For sine and/or cosine like $3\sin(4x)$ or $2\cos(4x)$ or $3\sin(4x) + 2\cos(4x)$, guess $A\sin(4x) + B\cos(4x)$, including both sine and cosine terms, even if $g(x)$ only includes sine or only includes cosine.

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If the right side of the differential equation is the sum or product of different types of functions, we need to multiply or add our guesses together, making sure that we have distinct constants, and that we've simplified the products of constants.

Here's a table of $g(x)$ functions, and the guess we should use for each one.

| $g(x)$ | **Guess** |
|---|---|
| $x^2+1$ | $Ax^2+Bx+C$ |
| $2x^3-3x+4$ | $Ax^3+Bx^2+Cx+D$ |
| $e^{3x}$ | $Ae^{3x}$ |
| $3\sin(4x)$ | $A\sin(4x)+B\cos(4x)$ |
| $2\cos(4x)$ | $A\sin(4x)+B\cos(4x)$ |
| $3\sin(4x)+2\cos(4x)$ | $A\sin(4x)+B\cos(4x)$ |
| $x^2+1+e^{3x}$ | $Ax^2+Bx+C+De^{3x}$ |
| $e^{3x}\cos(\pi x)$ | $Ae^{3x}(B\cos(\pi x)+C\sin(\pi x)) \rightarrow$ $e^{3x}(AB\cos(\pi x)+AC\sin(\pi x)) \rightarrow$ $e^{3x}(A\cos(\pi x)+B\sin(\pi x))$ |
| $(x^2+1)\cos(-2x)$ | $(Ax^2+Bx+C)(D\sin(-2x)+E\cos(-2x)) \rightarrow$ $(ADx^2+BDx+CD)\sin(-2x)+(AEx^2+BEx+CE)\cos(-2x) \rightarrow$ $(Ax^2+Bx+C)\sin(-2x)+(Dx^2+Ex+F)\cos(-2x)$ |

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Once we have a guess for our particular solution, before we can move on we need to make sure that none of the terms in the guess overlap with any terms in the complementary solution. For any overlapping terms, we'll need to multiply that section of the guess by $x$ (or maybe $x^2$, $x^3$, etc.) in order to eliminate the overlap.

For example, if the complementary solution includes the term $e^{3x}$, and our guess for the particular solution includes $Ae^{3x}$, then we'll need to multiply $Ae^{3x}$ by $x$, changing the guess to $Axe^{3x}$. For exponential terms like these, an overlap only exists if the exponents match exactly. So $e^{3x}$ and $Ae^{3x}$ are overlapping, but $e^{3x}$ and $Ae^{5x}$ are not.

Once we've eliminated any overlap and finalized the guess for our particular solution, we'll find the first and second derivatives of our guess, then plug those first and second derivatives into the original differential equation for $y'(x)$ and $y''(x)$, respectively.

Then we'll be able to combine like terms and equate coefficients on both sides in order to solve for the constants $A$, $B$, $C$, etc. Ultimately, we should end up with a particular solution that we can combine with the complementary solution in order to get the general solution to the second order nonhomogeneous equation.

Here's a summary of these steps.

1. Substitute $g(x) = 0$ to rewrite the nonhomogeneous equation as a homogeneous equation.

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2. Use the associated characteristic equation to solve the homogeneous equation, generating the complementary solution $y_c(x)$.

3. Make a guess for a particular solution.

4. Fix any overlap between the guess for the particular solution, and the complementary solution from step 2.

5. Find the first and second derivatives of the non-overlapping guess for the particular solution.

6. Substitute the derivatives into the original differential equation.

7. Equate coefficients to find the values of any constants.

8. Substitute the constant values into the guess to generate a particular solution $y_p(x)$.

9. Sum the complementary and particular solutions to get the general solution, $y(x) = y_c(x) + y_p(x)$.

Let's do an example so that we can see these steps in action.

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**Example**

Find the general solution to the differential equation.

$$y'' + 2y' = 4x - 6e^{-2x}$$

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We'll replace $g(x) = 4x - 6e^{-2x}$ with $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation (Step 1),

$$y'' + 2y' = 0$$

then we'll solve the associated characteristic equation.

$$r^2 + 2r = 0$$

$$r(r + 2) = 0$$

$$r = 0, \ -2$$

These are distinct real roots, so the complementary solution will be

$$y_c(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y_c(x) = c_1 e^{(0)x} + c_2 e^{-2x}$$

$$y_c(x) = c_1(1) + c_2 e^{-2x}$$

$$y_c(x) = c_1 + c_2 e^{-2x}$$ (Step 2)

Now we can switch over to working on a particular solution, $g(x) = 4x - 6e^{-2x}$. The first thing we notice is that we have a polynomial function, $4x$, and an exponential function, $-6e^{-2x}$. We'll use $Ax + B$ as our guess for the polynomial function, and we'll use $Ce^{-2x}$ as our guess for the exponential function. Putting these together (Step 3), our guess for a particular solution will be

$$y_p(x) = Ax + B + Ce^{-2x}$$

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Comparing this to the complementary solution, we can see that $c_2e^{-2x}$ from the complementary solution and $Ce^{-2x}$ from the particular solution are overlapping terms. To fix this (Step 4), we'll multiply $Ce^{-2x}$ from the particular solution by $x$, such that our guess becomes

$$y_p(x) = Ax + B + Cxe^{-2x}$$

Furthermore, we can see that $c_1$ from the complementary solution and $B$ from the particular solution are overlapping terms. To fix this overlap, we'll multiply $Ax + B$ from the particular solution by $x$, such that our guess becomes

$$y_p(x) = Ax^2 + Bx + Cxe^{-2x}$$

Taking the first and second derivatives of this guess (Step 5), we get

$$y_p'(x) = 2Ax + B + Ce^{-2x} - 2Cxe^{-2x}$$

$$y_p''(x) = 2A - 4Ce^{-2x} + 4Cxe^{-2x}$$

Plugging the first two derivatives into the original differential equation (Step 6), we get

$$2A - 4Ce^{-2x} + 4Cxe^{-2x} + 2(2Ax + B + Ce^{-2x} - 2Cxe^{-2x}) = 4x - 6e^{-2x}$$

$$2A - 4Ce^{-2x} + 4Cxe^{-2x} + 4Ax + 2B + 2Ce^{-2x} - 4Cxe^{-2x} = 4x - 6e^{-2x}$$

$$(2A + 2B) + 4Ax - 2Ce^{-2x} = 4x - 6e^{-2x}$$

$$(2A + 2B) + (4A)x + (-2C)e^{-2x} = (0) + (4)x + (-6)e^{-2x}$$

Equating coefficients from the left and right side (Step 7), we get

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$$2A + 2B = 0 \qquad 4A = 4 \qquad -2C = -6$$

$$2(1) + 2B = 0 \qquad A = 1 \qquad C = 3$$

$$2B = -2$$

$$B = -1$$

We'll plug the results into our guess for the particular solution (Step 8) to get

$$y_p(x) = (1)x^2 + (-1)x + (3)xe^{-2x}$$

$$y_p(x) = x^2 - x + 3xe^{-2x}$$

Putting this particular solution together with the complementary solution (Step 9) gives us the general solution to the differential equation.

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 + c_2e^{-2x} + x^2 - x + 3xe^{-2x}$$

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## When to use undetermined coefficients

This method of undetermined coefficients is most useful because it reduces the differential equation problem to an algebra problem. In this last example, we were able to get down to

$$(2A + 2B) + (4A)x + (-2C)e^{-2x} = (0) + (4)x + (-6)e^{-2x}$$

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and then solve for the constants with a simple set of equations.

The problem with the method is that it only works for a specific, somewhat small set of $g(x)$ functions. We'll typically use this method when $g(x)$ is some combination of polynomial, exponential, and/or trigonometric functions.

The next method we'll look at is variation of parameters, which is a little more involved than undetermined coefficients, but also more versatile, in the sense that it'll work for $g(x)$ functions that undetermined coefficients can't handle.

If we attempt to find the general solution with undetermined coefficients and it doesn't work (the system of equations we get from the algebra problem isn't solvable), then it usually means one of two things. It could mean that we made the wrong guess for the particular solution, and/or it could mean that undetermined coefficients can't be used to deal with the $g(x)$ in our differential equation.

Therefore, if we do have a problem finding a particular solution, we should first check for errors in our guess and the algebra steps we used with it. Then if that doesn't work, we should try variation of parameters instead, which we'll talk about in the next lesson.

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